

Orbit Guard - Technical Documentation

Team: RM99781
Public URL: <https://app-aegis-orbit-rm99781.azurewebsites.net/app/>
GitHub repository: <https://github.com/pebarone/aegis-orbit>
Primary SDG: ODS 9 - Industry, Innovation and Infrastructure

Solution and Space Context

Orbit Guard is a Space Situational Awareness MVP for satellite operators. It simulates deterministic orbital conjunction alerts, classifies collision risk from miss distance and collision probability, and returns evasion maneuver recommendations with ignition window, estimated delta-v, risk reduction, and residual collision probability. The solution connects to ODS 9 by supporting resilient space infrastructure, innovation, and operational reliability for satellite services.

Architecture

FastAPI + Uvicorn run on Azure Linux App Service. The React/Vite frontend is served by FastAPI under /app. API endpoints expose /health, /api/status, /api/conjunctions, /api/evasion-routing and Swagger at /docs. Azure Key Vault stores the solution secret, Managed Identity grants access, and Application Insights + Log Analytics monitor the App Service.

Requirement	Evidence
Public app	https://app-aegis-orbit-rm99781.azurewebsites.net/health returned status=healthy, service=Orbit Guard, version=0.3
Azure resources	Resource group rg-aegis-orbit-rm99781 contains App Service, App Service Plan, Key Vault, Application Insights, Log Analytics
Security	App Service httpsOnly=true, FTPS disabled, SystemAssigned Managed Identity enabled. Key Vault kv-aegis-rm99781
CI/CD	Workflow .github/workflows/deploy.yml triggers on push to main, uses actions/checkout, Azure login via secrets.AZURE_CREDENTIALS
Deployment evidence	Kudu/App Service deployment history shows multiple successful deployments, including OneDeploy records on 2023-07-20
Monitoring	Application Insights appi-aegis-orbit-rm99781 provisioningState=Succeeded. Alert rule alert-aegis-orbit-http5xx is enabled

Dashboard Evidence

Orbit Guard dashboard showing critical conjunction risk, simulated alerts, API health, orbital density matrix and event log.



Conjunction Feed Evidence

Conjunction feed showing active orbital events, severity, asset IDs, object IDs, TCA, miss distance and collision probability.

Orbit Guard Controle SSA

RELOGIO DA MISSÃO: 14:22:05 UTC [!] API AZURE: CONECTADA

OPERAÇÕES

NÓ SETOR-7G

PAINEL

CONJUNÇÕES

ROTA DE EVASÃO

STATUS DO SISTEMA

API NA MESMA ORIGEM

Lista de Conjunções

3 ATIVOS

Buscar ID...

STATUS	ID DO ATIVO	OBJETO ENVOLVIDO	TCA (UTC)	DIST. MIN. (KM)	PC
▲ CRÍTICO	AEO-LEO-104	DEBRIS-1998-067QZ	2026-06-03T14:35:00Z	0.420	3.10e-4
▲ ATENÇÃO	AEO-LEO-118	ROCKET-BODY-2015-021B	2026-06-04T02:10:00Z	1.900	8.00e-5
▲ NOMINAL	AEO-IOT-022	DEBRIS-2020-061AF	2026-06-04T19:45:00Z	4.600	1.50e-5

System Status Evidence

System status view showing the Orbit Guard service response, health endpoint and Azure readiness payload.

Orbit Guard Controle SSA

RELÓGIO DA MISSÃO: 14:22:05 UTC

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STATUS DO SISTEMA

Saúde da API e do Serviço

https://app-aegis-orbit-rm99781.azurewebsites.net

SAUDÁVEL

NOME DO SERVIÇO

Orbit Guard

STATUS DO DEPLOY

Versão

0.1.0

Equipe

RM99781

Alertas

3

/health + /api/status

```
{
  "health": {
    "status": "healthy",
    "service": "Orbit Guard",
    "version": "0.1.0"
  },
  "status": {
    "service": "Orbit Guard",
    "version": "0.1.0",
    "team": "RM99781",
    "mission": "Space Situational Awareness for orbital conjunction monitoring",
    "ods": "ODS 9 - Industry, Innovation and Infrastructure",
    "config": {
      "mock_token_secret_name": "SSA-API-MOCK-TOKEN",
      "key_vault_configured": true,
      "application_insights_configured": true,
      "external_api_mode": "disabled-deterministic-simulation"
    }
  },
  "azure_readiness": {
    "app_service_ready": true,
    "health_endpoint": "/health",
    "startup_command": "startup.sh",
    "managed_identity_expected": true
  },
  "simulated_alert_count": 3
}
```

API NA MESMA ORIGEM

Swagger Evidence

Swagger UI exposing health, status, conjunction and evasion-routing endpoints for the Orbit Guard SSA API.

Orbit Guard SSA API 0.1.0 OAS 3.1

/openapi.json

Deterministic Space Situational Awareness MVP for orbital conjunction monitoring and evasion routing.

default

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GET

/health

Health

▼

GET

/api/status

Status

▼

GET

/api/conjunctions

Conjunctions

▼

POST

/api/evasion-routing

Evasion Routing

▼

Schemas

^

EvasionRoutingRequest >

Expand all

object

HTTPValidationError >

Expand all

object

ManeuverRecommendation >

Expand all

object

ValidationError >

Expand all

object

Validation Summary

Validation	Result
Backend tests	python -m pytest: 8 passed.
Frontend build	npm run build --prefix frontend: TypeScript build and Vite production build succeeded.
Public smoke	/health returned healthy Orbit Guard 0.1.0; /app returned HTTP 200.
GitHub Actions visibility	Repository is public and workflow runs are visible through the GitHub API. Latest run 27042115968 completed successfully.
Known limitation	Azure Deployment Center sourcecontrols metadata is not currently linked as isGitHubAction=true because Azure Cloud Shell does not support the az deployment center sourcecontrols link command.

Command to connect Deployment Center metadata when a GitHub token is available:

```
az webapp deployment github-actions add --repo pebarone/aegis-orbit --branch main --resource-group rg-aegis-orbit-rm99781 --name app-aegis-orbit-rm99781 --runtime "PYTHON|3.12" --token <GITHUB_PAT>
```